Amcrest AD410 Doorbell

guide

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flyinglow 1 November 16, 2022, 6:56pm

I have integrated my AD410 doorbell into Home Assistant using the built-in amcrest integration and this **helpful comment** which pointed to @calisro 's **bug comment with a workaround** for problems with reliability in the Dahua integration.

I now have an entity for the doorbell in Home Assistant with sensors for online, doorbell pressed, human detected, and motion detected.

Here's exactly what I did (I'm still learning Home Assistant, so please let me know if there's something I should improve).

configuration.yaml

```
homeassistant:
  packages: !include_dir_named packages

amcrest: !include amcrest.yaml
```

secrets.yaml

```
amcrest_password: "PUT YOUR AMCREST AD410 PASSWORD HERE"
```

amcrest.yaml

```
- host: 192.168.XX.XX <-- PUT YOUR AD410 IP HERE
  username: admin
  password: !secret amcrest_password
  name: Front Door Bell
  resolution: high
  stream_source: rtsp
  scan_interval: 15
  binary_sensors:
    - online
```

Create a directory called `packages` under the directory that has your `configuration.yaml`. Put this file in the `packages` directory.

packages/package_amcrest_triggers.yaml

```
template:
  - trigger:
      - platform: event
        event_type: amcrest
        id: start
        event_data:
          camera: "Front Door Bell"
          event: CrossRegionDetection
          payload:
            Code: CrossRegionDetection
            action: Start
            data:
              Action: Appear
              Object:
                ObjectType: Human
      - platform: event
        event_type: amcrest
        id: stop
        event_data:
          camera: "Front Door Bell"
          event: CrossRegionDetection
          payload:
            Code: CrossRegionDetection
            action: Stop
            data:
              Action: Appear
              Object:
                ObjectType: Human
```

10 Likes

Amcrest AD410 no changes to entities

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calisro (Rob) 2 November 17, 2022, 7:13pm

One nice note for people who have this camera and use the camera.snapshot functionality in HA or even the picture glance (which uses the snapshot too). The Amcrest is set to terrible snapshot resolution. THis will change it. DIdn't brick mine but i'm not responsible for any issues. 🙂

```
http://admin:[email protected]/cgi-bin/configManager.cgi?action=setConfig&Encode[
```

They will now by High Resolution. To put it back, I think this works:

```
http://admin:[email protected]/cgi-bin/configManager.cgi?action=setConfig&Encode[
```

Its honestly beyond me why they cripple the camera with this low resolution crap.

2 Likes

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tzeench (Andrew Stenzel) 3 December 14, 2022, 4:55pm

> flyinglow:
>
> `action`

@flyinglow .
Would you care to share how you discovered which entities/triggers were even available for the AD410? The **Amcrest Integration** page doesn't show entities/triggers like the human motion detection or doorbell ring.

I'm interested in this discovery process as I have other devices in my HA which don't show all the entities/triggers that I expected.

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flyinglow 4 December 14, 2022, 6:33pm

I find it's really helpful to go to Developer Tools in Home Assistant. Then go to Events and where it says "Listen to events" enter `state_changed` and then "Start Listening"

You'll see a bunch of events go by, but in that set of events are the ones you care about from your other devices. That should help you find out about what events are triggered. I used that plus some trial-and-error with my AD410 doorbell to figure out the right events and data.

1 Like

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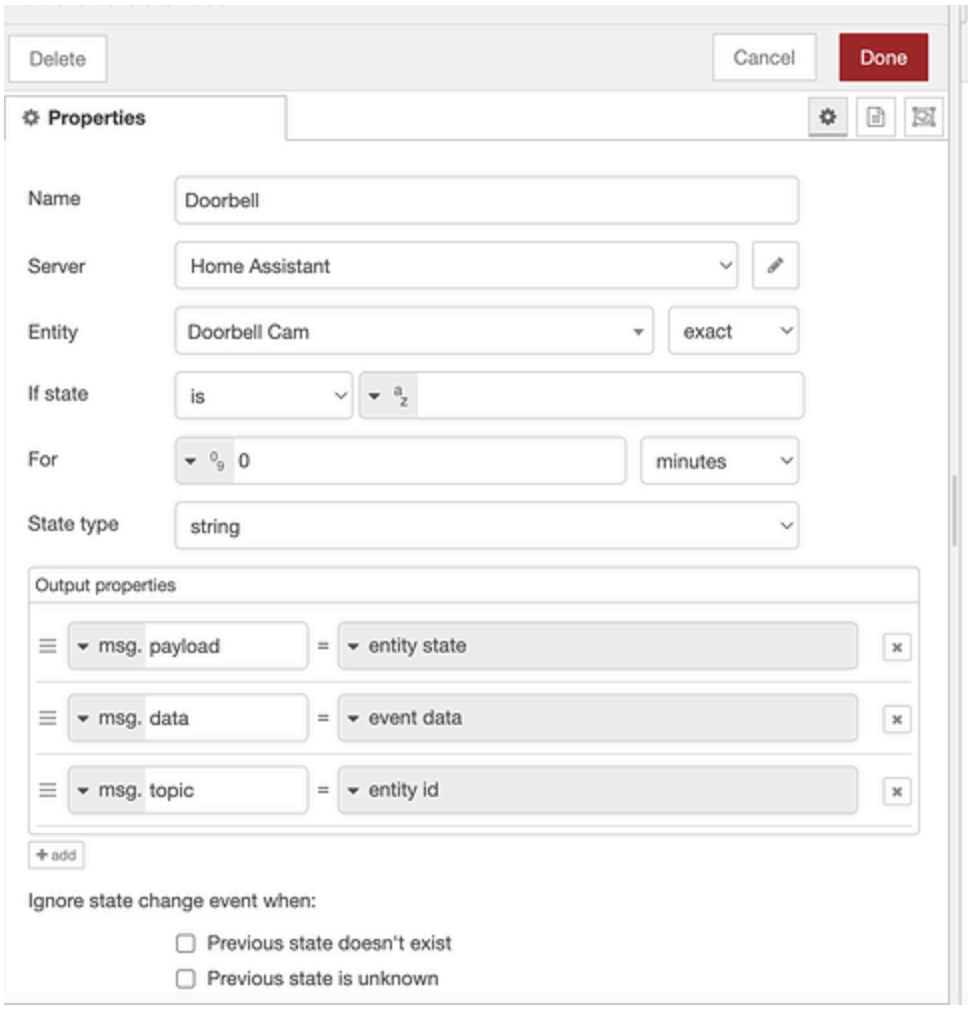
carefulcomputer 5 December 26, 2022, 7:15am

does it still work for you ?
I did exactly what you shared above. However, when I check event change node in node-red, I don't get any human detection or bell ringing state changes.



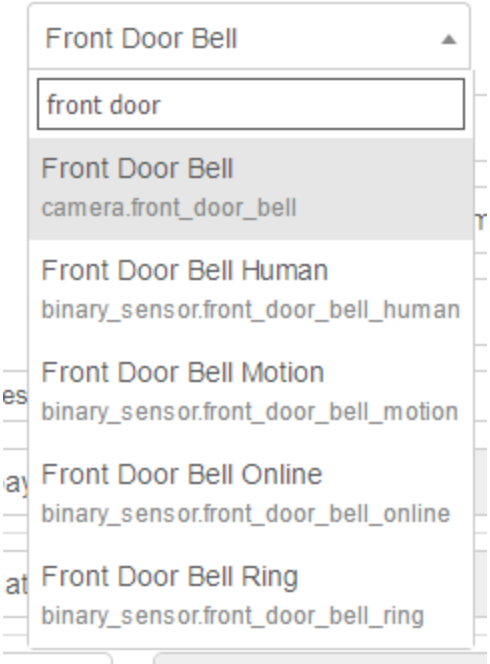
Only event I see is ‘streaming’. Could it be related to a breaking change in amcrest json ? ([Error on Human detected · Issue #67 · dchesterton/amcrest2mqtt · GitHub](#))
Or am I doing something wrong ?

[flyinglow](#) 6 December 26, 2022, 6:18pm

I think you’re choosing the wrong Entity in Node Red. It looks like you’ve selected the doorbell camera, which is the actual camera stream (which explains why the only event you see is ‘streaming’).

But there are 3 other entities that you should have if you followed what I shared above that correspond to the `binary_sensor` entries. In my example they are Front Door Bell Human, Front Door Bell Ring, and Front Door Bell Motion.

Expand the “entity” drop-down in Node Red and you should see the options to choose the other sensors. Here’s what I see:



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carefulcomputer 7 December 26, 2022, 7:58pm

yes !! my mistake. it works perfectly now. Thanks for putting this together !

Sagi-bb (Sagi Bb) 8 January 4, 2023, 4:16pm

This doesn't work for me , i am trying to send me a notification and nothing (also nothing when trying to open in browser)

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calisro (Rob) 9 January 10, 2023, 6:44pm

But that post you replied to wasn't about sending a notification. It was about changing the resolution of the camera.

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ian_bush (Ian) 10 January 18, 2023, 11:47am

> flyinglow:
>
> ```
> template:
>  - trigger:
>  - platform: event
>  event_type: amcrest
>  id: start
>  event_data:
>  camera: "Front Door Bell"
>  event: CrossRegionDetection
>  payload:
>  Code: CrossRegionDetection
>  action: Start
>  data:
>  Action: Appear
>  Object:
>  ObjectType: Human
>  - platform: event
>  event_type: amcrest
>  id: stop
>  event_data:
>  camera: "Front Door Bell"
>  event: CrossRegionDetection
>  payload:
>  Code: CrossRegionDetection
>  action: Stop
>  data:
>  Action: Appear
>  Object:
>  ObjectType: Human
> ```

First of all many thanks for this! It’s got my up and running with the AD410 nicely.

For me the human detection was not working. It looks like the ‘Action’ in the ‘data’ part should be under ‘Object’. An API change perhaps? So this works for me:

```
template:
  - trigger:
      - platform: event
        event_type: amcrest
        id: start
        event_data:
          camera: "Front Door Bell"
          event: CrossRegionDetection
          payload:
            Code: CrossRegionDetection
            action: Start
            data:
              Object:
```

```
                Action: Appear
                ObjectType: Human
      - platform: event
        event_type: amcrest
        id: stop
        event_data:
          camera: "Front Door Bell"
          event: CrossRegionDetection
          payload:
            Code: CrossRegionDetection
            action: Stop
            data:
              Object:
                Action: Appear
```

---

flyinglow 11 January 18, 2023, 7:05pm

Interesting. I wonder why mine is different. I just confirmed that my human detection is working with the original template I posted.

Maybe different firmware on our doorbells? From the “Firmware update” screen on the Amcrest App, mine is version 1.000.000000007.R.210220 (I may not have counted all the zeros correctly) and it says there are no updates available.

Thanks for figuring out how to make it work for you and posting the solution that’s working for you.

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ian_bush (Ian) 12 January 25, 2023, 11:35am

How odd, I am on the same fimrware version of the doorbell, and if I understand correctly this info is coming straight from the doorbell so strictly local. Only other thing I can think of is some different settings could be causing it.

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silentjon (Jon) 13 January 25, 2023, 6:36pm

I had to use Ian’s as well, and I’m getting human notifications now as well, same firmware.

Is it possible to switch the doorbell to H264? I’m not seeing a setting inside of the amcrest app for that. I changed my other amcrest cameras from H265 to H264 and it fixed the issue I’m having with the doorbell.

Basically when I open it in homeassistant/chrome browser… I get the error that keeps popping up/disappearing in about 0.1 seconds that says “Error with media stream contents”

I setup a human detection automation that attempts to stream the camera to a google hub, but it never works. Really I just want to get it to stream somehow to an android tablet, how did you all go about some sort of automation for that?

GaryOkie 14 January 25, 2023, 7:02pm

@silentjon - the AD410’s factory default encoding is H.264 - at least it was with the original 20210220 firmware.

Yes, it can be changed in several ways. Via the HTTP API, Amcrest/Dahua NVR UI, or Amcrest Surveillance Pro & related s/w.

You can confirm what your primary stream encoding is via this API command entered into a browser:

```
<ip_address>/cgi-bin/configManager.cgi?
action=getConfig&name=Encode[0].MainFormat[0]
```

Edit: Be advised that changing video settings using software other than SmartHome app can sometimes introduce undesirable artifacts. It may change just the encoding and all is well, or it can add additional crap the AD410 doesn’t like. May be safest to just use the API.

Here’s the API command to change the main stream encoding:

```
<ip_address>/cgi-bin/configManager.cgi?
action=setConfig&Encode[0].MainFormat[0].Video.Compression=H.264
```

1 Like

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silentjon (Jon) 15 January 25, 2023, 7:47pm

Awesome, it was at H.265 for some reason, I used that API link and it’s working now for me. A little buffery, but probably due to the wifi connection. The error isn’t popping anymore now.

I’ve went through locking up my doorbell many times from using Amcrest Windows tools and having to reset it, so I’m glad these API links work successfully.

1 Like

drew.hansen (Drew Hansen) 16 February 2, 2023, 9:08pm

Just wanted to drop a note saying thanks for this! I tried using the Dahua integration which was super unreliable. This is working great and my wife is now happy 🙂.

1 Like

Gorgonito (The gorgonator) 17 February 6, 2023, 11:25pm

Stopped by to say thank you for your work! 😃 I had been trying to make this doorbell to work consistently and the Dahua integration I had been using wasn't reliable at all.

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adoucette (Ari) 18 March 5, 2023, 8:16pm

Thank you for helping me set up the AD410 in Home Assistant. How would I then use this in an automation through HA's UI when a human is detected by the doorbell camera? I see that I can select event triggers when I build an automation through HA's UI, but don't know what to do from there.
Thanks for any assistance.
Ari

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flyinglow 19 March 5, 2023, 9:31pm

I think this guide will help you get pointed in the right direction
Automating Home Assistant - Home Assistant

What would you like to happen when a human is detected by the doorbell camera?

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adoucette (Ari) 20 March 6, 2023, 1:06am

Thank you - I've previously used that guide and felt it was indeed a good intro.
I'm not seeing the "Front Door Bell Human" event in Home Assistant. The Amcrest app notices humans and sends me notifications, but HA doesn't pick up the event from the config above.

Note I've tried both @flyinglow the OP's code and that from @ian_bush
As Iook back through the logs from the Amcrest integration, I don't see any events getting logged except that the "online" binary sensor is constantly cycling off to on and back again.
I'd like to find out how to get the human event working and then find out how to enter it properly into the automations GUI to use as a trigger.

GaryOkie 21 March 6, 2023, 4:17pm

Have you updated your AD410 firmware to 20221201 ? If so, that explains why you aren't seeing the CrossRegionDetection (Human detected) event. In fact, no events are accessible with this corrupted firmware update. Amcrest developers admitted to the exact same mistake with the AD110 firmware which took a few months to correct. ~~No word yet on when/if this AD410 firmware will be corrected.~~ EDIT: The latest firmware on the download page has been updated to fix this issue.

Note that within the SmartHome app it still shows xx210220 as the production version. It does not reflect or notify that there is a newer firmware available.

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adoucette (Ari) 22 March 14, 2023, 10:17pm

You were correct @GaryOkie . On the 20221201 firmware the human detection was not registering and the doorbell was disconnecting and reconnecting to wifi a few times every minute. After downgrading to the previous firmware both problems went away.

GaryOkie 23 March 14, 2023, 11:01pm

@adoucette Glad to hear you've resolved the event problem by reverting the firmware. Hope your WiFi connection remains stable as I had not heard of that being an issue with the new firmware.

Speaking of which, there is a newer firmware available now which the change log (Yea! finally!) states the event handling issue has been corrected. I have not installed it yet. If you do, please report back if you have any issues. You still have the old 20210220 firmware that you can go back to if needed.

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derekjwhitten (Derek) 24 March 15, 2023, 7:14am

Can anyone comment on how quickly the events fire using this set up? I'm currently using the Dahua integration but find doorbell press and human motion detection take up to 10 seconds to fire through to Home Assistant. I'm looking for a way to improve this and would like to hear any feedback before I try setting up using this approach.

XcOM9876 26 March 16, 2023, 5:00pm

Using the info I learned here, I am monitoring for events that are amcrest and filtering them for the doorbell button press, I get notified almost instantly using node red to process it and send the notification with a snapshot

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GaryOkie 27 March 16, 2023, 5:10pm

"events that are amcrest" are from the core Amcrest integration. Derek was questioning why events from the Dahua integration are taking up to 10 seconds. They should be instantaneous as well, but I can't confirm as I'm back to using the Amcrest core integration for a while now.

XcOM9876 28 March 16, 2023, 7:39pm

Ahh my bad, that's what I get for not reading lol

BigWalnutZ 29 March 19, 2023, 3:47pm

I think you read it properly. @derekjwhitten was asking:

> derekjwhitten:
>
> how quickly the events fire using this set up?

…which I interpret to be he is comparing his slow 10 s Dahua setup to what we are experiencing with the Amcrest integration and the events triggers to provide info as to whether he would want to try configuring this way instead.

That being said, as you said, I too get notified on my watch immediately the moment I press the button and similarly when it detects a human. For testing it configured an automation to turn on a light switch when the doorbell button is pressed and it is simply like turning on a light switch it is that quick, even a split second before I receive the push notifications. I haven't configured mine to do the snapshot yet so I can't speak to that portion of it. (still on test bench, not installed yet)

All around good work with this @flyinglow . I would hope we can bring this to the attention of the Amcrest integration owner so they possibly add these in as part of their documentation or better yet use this to create the actual sensors and whatnot out of the box so we don't have to use your technique. Also the using the `AlarmLocal` is so much better for me with 2 doorbells, that I can identify which doorbell triggers the automation.

This is my test automation to see if I could properly determine which doorbell was pressed. My triggers add looking for the `camera: <name of camera>` and setting different trigger IDs to be used in the `choose` action down the automation. I also have an ESPHome device w/ 2-channel relay configured to be able to momentary press my mechanical doorbell chime as the chime kit is not compatible with mine:

```
- id: '1674677926052'
  alias: Amcrest Doorbell Pressed
  description: Trigger when Amcrest Doorbell Button Press Event Fires
  trigger:
  - platform: event
    event_type: amcrest
    event_data:
      camera: Garage Doorbell
      event: AlarmLocal
      payload:
        action: Start
    alias: Amcrest Garage Doorbell Pressed
    id: Garage Doorbell Pressed
  - platform: event
    event_type: amcrest
    event_data:
      camera: Front Doorbell
      event: AlarmLocal
      payload:
        action: Start
    alias: Amcrest Front Doorbell Pressed
    id: Front Doorbell Pressed
  action:
  - choose:
    alias: Determine which doorbell was pressed and perform actions accordingly
    - conditions:
      - condition: trigger
        id: Garage Doorbell Pressed
```

Confirming this works, I now have modified the `packages\amcrest.yaml` to have presses and sensors for each, but I'll just show the different doorbell button presses here:

```
  - trigger:
      - platform: event
        event_type: amcrest
        id: start
        event_data:
          camera: "Front Doorbell"
          event: AlarmLocal
          payload:
            Code: AlarmLocal
            action: Start
      - platform: event
        event_type: amcrest
        id: stop
        event_data:
          camera: "Front Doorbell"
          event: AlarmLocal
          payload:
            Code: AlarmLocal
            action: Stop
    binary_sensor:
      - name: Front Doorbell Ring
        icon: mdi:doorbell-video
        state: >-
          {{ 'on' if trigger.id == 'start' else 'off' }}

  - trigger:
      - platform: event
        event_type: amcrest
```

So, I will swap out the original test triggers for that automation and just use the binary sensors. This should work, but haven't tested yet.

```
- id: '1679240460251'
  alias: Amcrest Doorbell Pressed (New triggers)
  description: Trigger when Amcrest Button Press Event Fires
  trigger:
  - platform: state
    entity_id:
    - binary_sensor.front_doorbell_ring
    to: 'on'
    id: Front Doorbell Pressed
  - platform: state
    entity_id:
    - binary_sensor.garage_doorbell_ring
    to: 'on'
    id: Garage Doorbell Pressed
...
```

Sorry this got long, but hope it helps someone else that may have similar situations with having more than one doorbell.

1 Like

GaryOkie 30 March 19, 2023, 5:36pm

I thought Derek asking about “this setup” meant his setup using the Dahua Integration which is slow… not the OP setup.

I concur that the Amcrest Integration provides immediate event triggers, and would greatly benefit from adding new sensors. There have been PR’s to do just that. These PR’s have been rejected due to the NO MORE YAML CONFIG rule. So until the effort is undertaken to modernize this integration to use the config flow UI, we need to rely on parsing events. Ain’t that hard, and examples like yours are indeed helpful to new users.

---

cmooty73 (Cmooty73) 31 April 2, 2023, 10:26pm

Post edited.
Guys, I struggled getting template sensors to work, turns out they defaulted to unknown, and never came online until I went out and tripped them. I just was assuming they’d default to off. I’m a bone head.

---

derekjwhitten (Derek) 32 April 6, 2023, 7:59am

Apologies everyone for the ambiguity in my post…

I was asking about any latency observed with the Amcrest integration and the tips shared by @flyinglow . I’m currently using the Dahua integration and find the Amcrest app goes off on my phone almost instantly, but the event in HomeAssistant is about 10 seconds behind. I’m running into further issues with my set-up since my early post, so I’ll have a go at flyinglow’s approach this weekend and see how it goes.

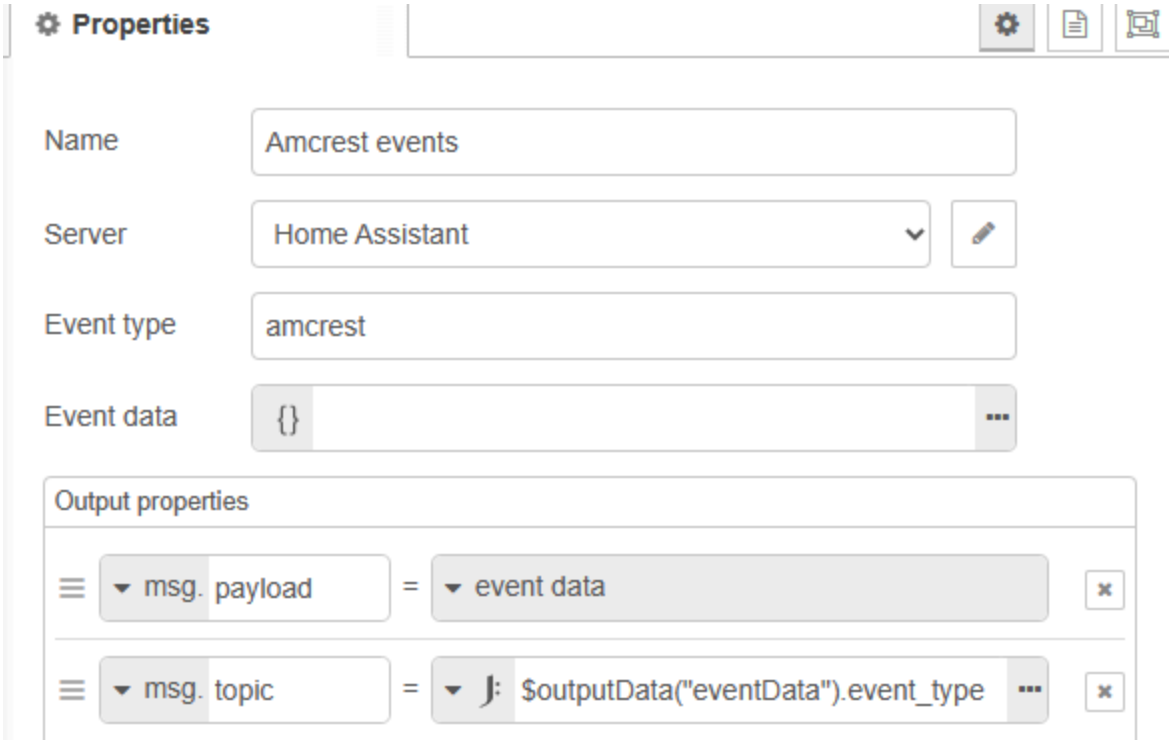
XcOM9876 33 April 18, 2023, 9:27pm

What is your setup you are using to run HA? are any other integration’s slow?

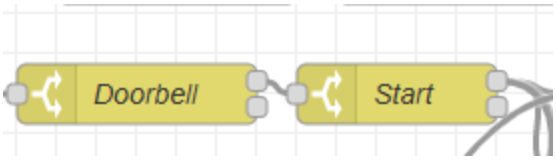
As @BigWalnutZ does, I also check events, depending on how you are filtering them this can be slow, if you pull in all events and then filter it out after looking for it this is a slow process and hammers HA, the

better option if this is the case is to filter events only for what you are looking for from the off.

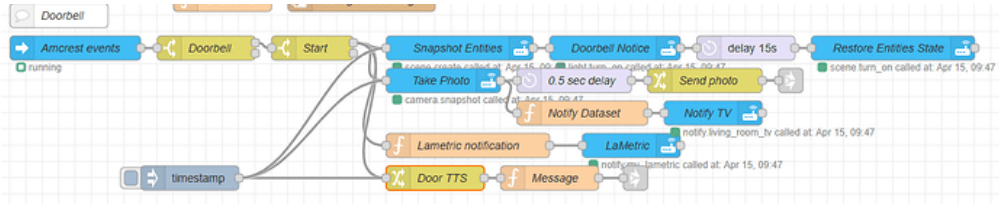
So my system is filtered from the off to only look at Amcrest events:



Where after it looks for AlarmLocal and Start



And then runs the rest of the automation



Any further info you can give about your setup will help, could be a network issue, HA issue, integration issue.

1 Like

Diceman1 34 April 22, 2023, 9:46pm

I did all this and I can see the camera display, and when i press my doorbell I can play a sound on all my google minis, but then afterwards the Front Door Bell Ring is stuck on “On”. I have to manually go turn it

“Off” in the developer tools each time, Any idea why this may be? The person detection also just triggers On, and then never back off as well.

Thanks.

---

pexeuskt 35 May 10, 2023, 7:20am

I’m actually stuck and found this post after trying that exact thing (walk outside and push the button then show my face). I’m certain both events worked within the camera because the native Amcrest app gave me the relevant event alerts.

Firmware is 20210220 and the “doorbell Online” sensor shows connected, the streaming picture works, but the Human, Motion, and Ring sensors all show unknown. Over in the event viewer I subscribed to event type “amcrest” and I see plenty of results in therms of motion and video recording, but I haven’t see my button presses or human detection show up.

I wonder if that’s enough info to trigger any other ideas on your part?

GaryOkie 36 May 10, 2023, 2:29pm

It sounds like your integration is working fine using the event type “Amcrest” since you are seeing all the events showing up.

But… " *Human, Motion, and Ring sensors all show unknown*"? The core Amcrest integration doesn’t have a doorbell ring sensor! It has a motion sensor and a Crossline Detection (tripwire) sensor, but not a human motion detected sensor (aka CrossRegionDetection).

You can craft your own sensors using events or just trigger on these events directly.

Maye you have confused the custom Dahua integration configuration which provides a plethora selection of sensors that the Amcrest integration does not.

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pexeuskt 37 May 10, 2023, 4:15pm

That could very well be true - I’m a bit confused by this thread and I’m pretty much copying the YAML Here’s what I’ve got right now:

```
# amcrest AD410 doorbell
amcrest:
  - host: 192.168.1.12
    username: admin
    password: [redacted]
    name: frntdoorbell
    resolution: high
    stream_source: rtsp
    scan_interval: 15
    binary_sensors:
      - online

template:
  - trigger:
      - platform: event
        event_type: amcrest
        id: start
        event_data:
          camera: "Front Door Bell"
          event: CrossRegionDetection
          payload:
            Code: CrossRegionDetection
            action: Start
            data:
              Action: Appear
              Object:
                ObjectType: Human
      - platform: event
```

That's using the `amcrest` integration and creating the binary sensors that should do their obvious things, no? (meaning that the "human" sensor goes ON when the AD410 notices a person, etc etc). The sensors show in my default dashboard, they're just always in the `unknown` state.

GaryOkie 40 May 10, 2023, 7:41pm

I don't use the techniques described in this thread that I see you have closely followed. There are some things that just look really weird to me unless firmware differences are behind it.

For example, the Doorbell press event isn't "AlarmLocal". That event is associated to the older AD110 doorbell for motion events detected via its PIR sensor. The correct doorbell press event is "CallNoAnswered".

Both the AD110 and AD410 report "VideoMotion" events for pixel-change motion but typically you should disable Motion Detection for the AD410 in the SmartHome app as it throws too many false motion events. Just opt to use only Human Detection in the AD410 and check for the CrossRegionDetection event.

I use a much simpler method of triggering on AD410 events that doesn't rely on creating binary sensors as the examples in this thread show.

Automation.yaml

```
- id: doorbell_button_pressed
  alias: Doorbell button pressed
  trigger:
  - platform: event
    event_type: amcrest
    id: DoorbellPress
    event_data:
      event: CallNoAnswered
      payload:
        action: Start

  action:
    - service: script.doorbell_press_actions


- id:    human_motion_detected
  alias: Human Motion Detected
  mode: parallel
  max: 5
  trigger:
  - platform: event
    event_type: amcrest
    id: Human Motion
    event_data:
      event: CrossRegionDetection
      payload:
        action: Start
        data:
          ObjectType: Human
  action:
    - service: script.human_detected_actions
```

---

pexeuskt 41 May 14, 2023, 5:15am

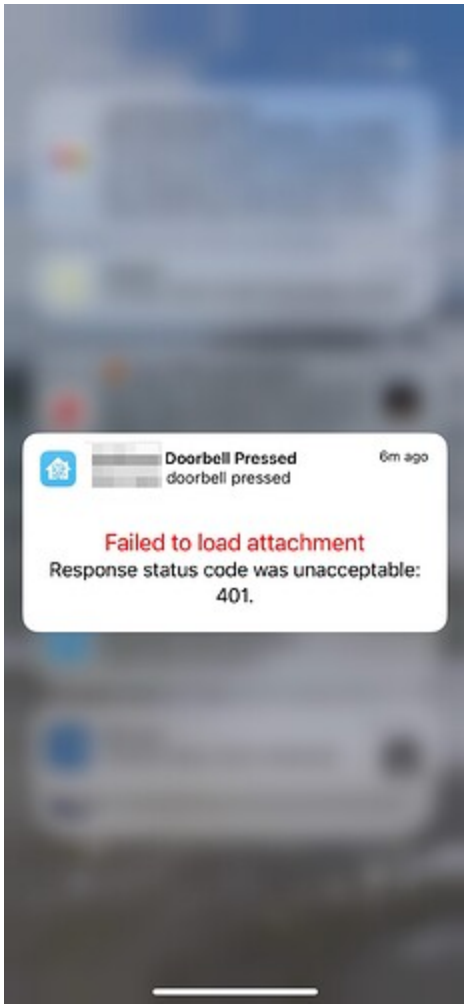
awesome - this is working much better (obviously after making sure the action pointed to something real, since I didn't have a 'script.human_detected_actions' setup yet… but I will soon).

Good stuff, thx!

1 Like

pexeuskt 42 May 14, 2023, 7:56pm

Well, I'm making serious progress but I'm unable to get snapshot images to come through.




```
- id: doorbell_button_pressed
  alias: Doorbell button pressed
  trigger:
  - platform: event
    event_type: amcrest
    id: DoorbellPress
    event_data:
      event: CallNoAnswered
      payload:
        action: Start
  action:
  - service: notify.notify
    data:
      title: xxxxx Doorbell Pressed
      message: xxxxx doorbell pressed
      data:
        image: http://192.168.1.26/cgi-bin/snapshot.cgi
        authentication: digest
```

```
        username: admin
        password: [redacted]
```

anything obviously wrong here?

GaryOkie 43 May 14, 2023, 10:27pm

I assume you are using the HA Companion Notifier? I've not used that before so can't really say why it isn't liking the snapshot.cgi URL. (Digest auth is correct).

What I do is call the camera.snapshot service to create a jpeg then send that image file using Telegram notifier. I believe you can do the same with Companion notifier instead of a URL.

Edit: maybe the problem is you specified image: for the url, instead of "url:"? The error message suggests that as well (failed to load attachment).

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pexeuskt 44 May 15, 2023, 6:12am

> GaryOkie:
>
> I assume you are using the HA Companion Notifier? I've not used that before so can't really say why it isn't liking the snapshot.cgi URL. (Digest auth is correct).

I mean, I'm not completely sure but I think the answer is "yes" given I'm pretty much just trying to implement what I found here: **Standard Attachments | Home Assistant Companion Docs**

I spent the last hour or so playing around trying to do things with two separate actions: save a temp image then directly attach it, but I've pretty much hit a wall trying to figure out the (probably pretty basic aspects) of doing that. oh well ¯_(ツ)_/¯

GaryOkie 45 May 15, 2023, 12:15pm

Probably best to ask for help with notifications in the Companion thread. Just took a quick glance at the doc and see the URL needs to be internet-accessible. Because of that requirement you should definitely focus on using a snapshot jpg as an attachment instead.

pexeuskt 46 May 16, 2023, 4:15am

Looks like **Standard Attachments | Home Assistant Companion Docs** was very relevant (voodoo requiring me to add `/local/` in the middle of a path when trying to attach an image, which was necessary and did work). To your point it's about "Internet" accessible (which I think includes just plain LAN) as opposed to "filesystem" accessible.

I ended up tacking in another direction after I learned I can reference the camera entity directly in the notification.

Here's the complete YAML. This intentionally notifies all mobile devices (hence `notify.notify` as opposed to a specific mobile device). This displays a static thumbnail and after the correct amount of clicking it gets to the live stream for this device.

```
- id: doorbell_button_pressed
  alias: Doorbell button pressed
  trigger:
  - platform: event
    event_type: amcrest
    id: DoorbellPress
    event_data:
      event: CallNoAnswered
      payload:
        action: Start
  action:
    service: notify.notify
    data:
      message: 'Ring pressed at {{ now().strftime("%Y-%m-%d %H:%M:%S") }}'
      data:
        entity_id: camera.ENTITYNAME
      title: 'Doorbell'
```

I have things to adjust or improve here:

1. I think these are literal LAN subscribed alerts only, which isn't good enough. I want these to work seamlessly home or not. I think that's why folks suggest using various third party services.
2. The number of clicks required to get to the video feed is, well, *more than one* so I want to optimize that.
3. I'd love to get proper 2way audio working.

GaryOkie 47 May 16, 2023, 8:21pm

2-way talk in HA is now possible, but it may be a fairly large rabbit hole to drop into. You're looking at having to merge a number of additional components to put it all together. Frigate, Go2RTC to name a few.

Start with this Youtube video to check it out… **Amcrest Smart Doorbell w/ Home Assistant & Two-Way Audio - YouTube**

---

pexeuskt 48 May 16, 2023, 11:52pm

Noted! I'm also toying with Scrypted as an HA add on, which I hear has great support for doorbells if you want to engage with them via iOS ecosystem (which happens to be the case for me at the moment… but mobile-os-agnostic is still important).

---

omayhemo (Doug) 49 May 20, 2023, 7:35pm

Thought I'd drop this here, because OMG!

My AD410 has some difficulty staying connected to WiFi. The AP is less than 20' away, which makes this odd to begin with. What's worse, is when it drops WiFi it attempts to reconnect with spoofed MAC addresses, until the unit is power cycled. This is a problem because I reserve addresses on my network and MAC filter.

I contacted Amcrest about this behavior for a solution, hoping to have an option to turn off hardware address spoofing. The answer I received is mind-numbingly obtuse.

After inquiring about my network, they chose part of the topology on which to place the blame. In this case Ubiquiti UniFi APs. Claiming that this is a problem with UniFi that they've heard about before and cannot solve themselves. MMMMk, so pfsense is actually where the MAC is registered, and the IP address reserved. There is no value in UniFi nor pfsense in wrecking device registration by spoofing their hardware addresses. Not that that is even a thing.

They didn't even think through this answer before setting forth the lie. My frustration with Amcrest has widened to the point where I'm done with them. Their software (including firmware) is an abomination, and their hardware is from Sino manufacturers. And now this kind of service.

---

parautenbach (Pieter Rautenbach) 50 May 20, 2023, 9:16pm

Sorry to hear about your frustrations, but I have an AD110 that works flawlessly.

omayhemo (Doug) 51 May 20, 2023, 10:42pm

Do you use RTSP?

---

parautenbach (Pieter Rautenbach) 52 May 21, 2023, 6:14am

I use the built-in Amcrest integration with the default of `mjpeg`. While I have other cameras (not Amcrest) that uses RTSP, I had no need to set it as it was working fine.

---

omayhemo (Doug) 53 May 21, 2023, 1:28pm

Interesting, not an option for as I use DW Spectrum. None the less, the point of my post was the incredulous claims support was making, to claim MAC spoofing is occurring at the AP.

DeltaNu1142 54 June 9, 2023, 2:32pm

I generally keep my AD410 isolated from the Internet, but the picture was all washed out and I wanted to see if I could change any settings in the app… so I allowed it to connect to the Internet so I could log into it. A firmware update popped up and, without thinking, I clicked “update.” Fortunately, everything still seems to be working, but it required the Dahua integration to be restarted.

Old version: 1.000.0000000.7.E.210220
New version: 1.000.00AC001.0.R.20230221

Bartem 55 June 20, 2023, 11:05am

I went back and forth with Amcrest Support for weeks with them making the same ridiculous claims about my network. My issue was I could not link the doorbell to my Amcrest NVR at all, it worked fine in every other aspect. Ultimately updating the firmware to the latest and changing two fields in their Amcrest IP Config Utility or Amcrest Surveillance Pro app, my issue was resolved. The “Issues” with my network occurred with all 3 RMA’d doorbells (Also had the lens flare issue with IR on)… Anyway my post is the

last one on this thread: **New Firmware for AD110 and AD410 Available - Page 7 - Amcrest Forum**
Changing the IP and Subnet mask (but only with updated firmware) resolved all of my network issues, hopefully it will help you as well.

1 Like

---

omayhemo (Doug) 56 June 26, 2023, 4:22pm

I have the latest. Simply put, Amcrest seems to suck, unfortunately. what's even better is it appears the operating temperature is incapable of withstanding a US South Summer. Went tits up yesterday at 100 degrees F.

ChileHA 57 July 27, 2023, 8:25pm

Hey everyone, I only get unknown States on all the entities. The firmaware is: 1.000.00AC001.0.R.20230221

my configuration.yaml is:

```
amcrest:
  - host: 192.168.1.105
    username: !secret amcrest_username
    password: !secret amcrest_password
    name: Timbre Amcrest
    resolution: high
    stream_source: rtsp
    scan_interval: 15
    binary_sensors:
    - online
    switches:
      - privacy_mode

  - host: 192.168.1.105
    username: !secret amcrest_username
    password: !secret amcrest_password
    name: Timbre Amcrest sub
    resolution: low
    stream_source: rtsp
    scan_interval: 10
    binary_sensors:
    - online
    switches:
```

```
        - privacy_mode
```

here is my log:

```
Home Assistant Core
<Unconnected @ 192.168.1.105> Trying again due to error: ConnectTimeout('')
5:37:37 PM – (WARNING) components/amcrest/__init__.py - message first occurred at
amcrest: Error on device update!
July 26, 2023 at 10:04:27 PM – (ERROR) Switch - message first occurred at July 26
Error while setting up amcrest platform for camera
July 26, 2023 at 10:04:26 PM – (ERROR) Camera - message first occurred at July 26
Error while processing events from Timbre Amcrest sub camera: LoginError()
July 26, 2023 at 10:04:23 PM – (WARNING) Amcrest - message first occurred at July
Timbre Amcrest sub camera offline: Login error:
July 26, 2023 at 10:04:23 PM – (ERROR) Amcrest - message first occurred at July 2
```

What can I do???

GaryOkie 58 July 27, 2023, 11:58pm

Long time ago I tried setting up both a high and low res config for the same camera. I found it isn't supported - at least, not with the Amcrest integration. The integration gets confused in a number of ways when the same IP is used for multiple camera entities.

Now I'm not exactly sure that is the reason for your particular errors. There wasn't this privacy switch option back then. Try removing the lo res sub camera config and see if everything works OK.

ChileHA 59 July 28, 2023, 1:48am

Still the same problem. Note I also have Dahua integration. Could that cause a problem?

GaryOkie 60 July 28, 2023, 2:53pm

The Dahua integration should be able to coexist with the Amcrest integration, but ideally it would be sensible to use one or the other if possible. I'm not sure what would happen if you used 2 different integrations for the same camera IP, but theoretically I think it should work.

You should also review RRollers Github page for his Dahua custom integration for known issues.

ChileHA 61 August 30, 2023, 4:00pm

any feed back on the new firmware 1.000.00AC002.0.R.20230810?

GaryOkie 62 August 30, 2023, 4:24pm

I contacted Amcrest Support 2 weeks ago when this update first came out asking for full release notes. He replied that none were available to him but he would ask the dev team and post them online in the Amcrest forum. Nothing has been posted yet.

I also mentioned that my AD410 has been disconnecting/rebooting during very hot (>100F) days. He admitted that this has been a problem and for me to go check my doorbell to see if the bottom case has split open due to a known issue with its internal lithium battery swelling like a gas balloon. Fortunately, mine hasn't - but I have seen a number of reports online from others that have had this happen.

I also learned that the new firmware has tweaked how the AD410 charges its lithium battery to supposedly minimize damage from overheating. So that is the extent of what I know this firmware update provides. I haven't installed it yet, hoping like you, to learn more about it and to see if any Home Assistant users have had any issues with it.

3 Likes

---

thornygravy 63 September 1, 2023, 12:29am

Wow if only I had updated, I got the spicy pillow a couple weeks ago.

Good news is, if you remove it everything still works. Just mentioning for those who may run into this.

---

odwide 64 September 1, 2023, 3:20am

This is the changelog for fw 1.000.00AC001.0.R.20230221.

GaryOkie 65 September 1, 2023, 8:47pm

@thorngravy - I've heard if you don't have the chime kit installed you can remove the battery - no issue. But I sure thought that battery was needed when the doorbell button is pressed since the kit diverts all power to the chime. Please confirm if you have the chime kit installed or not. Thanks.

@odwide - yes, a brief change log was provided for the Feb 2023 release. There has not been anything published from Amcrest for the latest Aug 2023 firmware {20230810}.

Amcrest hasn't even got around to updating their firmware download site to include the latest version. It has to be installed via the SH app currently.

---

odwide 66 September 2, 2023, 2:34am

Got it. Didn't know there was a new fw from 8/10. I did install the one from February and the image looks smoother now.

---

thornygravy 67 September 4, 2023, 2:22am

@GaryOkie yea, no chime kit

GaryOkie 68 September 20, 2023, 9:04pm

I finally got a reply from Amcrest Support regarding the "missing" change log for the latest Aug 2023 AD410 firmware…

Here it is:

```
Change log:

Lexus_SDB61i-Lexus2_Eng_N_32M_CA_V1.000.00AC002.0.R.20230810.bin
08-10-2023

The update incorporates the following enhancements:
```

```
- Changed the charging threshold for an internal battery to resolve some
overheating issues.
- Updated the logging driver, closed the kernel write log function.
- Fixed a security scanning bug.
```

> I will see that this changelog is uploaded to our website today.

I've held off installing firmware updates on this doorbell, but this info leads me to believe it will be worthwhile - particularly since my AD410 has been shutting down/rebooting during this past summer's heat.

I probably have a partially swollen battery that will need to be replaced, but without opening the case, I won't know for sure. Now that the temps are normal, it has been working fine.

MT0321 69 September 21, 2023, 2:50am

I upgraded to 1.000.00AC002.0.R.20230810 a few days ago and haven't encountered any problems. I wasn't having any before either, though.

---

asanseau (Alex) 70 October 3, 2023, 11:20pm

Slightly off discussion I admit, but would anyone here know where to buy the Amcrest AD410 Doorbell from in the UK? Thanks in advance.

GaryOkie 71 October 4, 2023, 3:06pm

I'm in the US, so can't confirm - but according to a couple of responses in **this reddit thread**, you can order it from Amazon US and have it shipped to UK. They didn't mention shipping costs though.

You should also consider the Reolink Doorbell. It's available in the UK and its integration is well supported in HA.

Note: I'm still relatively happy with my AD410 & AD110 doorbells and have a lot of experience with their CGI API and Home Assistant integration. But if I ever need to replace them, the Reolink is what I would go for.

calisro (Rob) 72 October 8, 2023, 2:11pm

Anyone know where I can find this firmware version? I want to upgrade to the newest but want to make sure I can get back to my current one.

1.000.0000000.7.R.210220

GaryOkie 73 October 8, 2023, 2:30pm

Since this 210220 version isn't listed on the Amcrest firmware download site any longer, you will need to contact Amcrest Support to request it.

BTW - I'm still on that original version and haven't updated yet either. Please reply if they provide you with a "public" link you can share. I'm in no rush to update, but having the original version as a backup is a smart move.

1 Like

---

calisro (Rob) 74 October 10, 2023, 3:52pm

I have received the firmware but they send it through email. Happy to forward it to you if you like. Alternatively, if you contact them, they sent it to me in less than 30 minutes.

1 Like

---

calisro (Rob) 75 October 10, 2023, 6:29pm

And the FW update didn't seem to break anything in my automations/events. Yet. 😉

1 Like

---

thornygravy 76 October 16, 2023, 10:07am

thanks for posting your findings, useful information for everyone

calisro (Rob) 77 October 18, 2023, 1:43pm

welp. Upgraded my firmware to the latest Amcrest one. Seemed to work okay at first. But new firmware:

1. sending audio via the api to the doorbell no longer works(I am not talking 2-way audio. Just the tts/file to doorbell api).
2. Whenever the doorbell was rung, the amcrest integration went offline and I think the bell was restarting.

I've since rolled back. 🙂

GaryOkie 78 October 18, 2023, 2:13pm

Thanks Rob for reporting back. I have still held off updating, and will probably wait until just before next summer's heatwave to do so - IF the doorbell lasts that long. I am getting a few occasional reboots at random times, but they are not related to doorbell presses.

It sounds like your second issue with the new firmware might be related to the chime configuration. If the firmware is configured for no chime, it probably won't trigger the chime kit relay.

Did you do a factory reset per the change log instructions? I'm not sure this is necessary, but it may be needed to ensure the configuration is correct. Even so, that is unlikely to fix the TTS/File API issue. That will need to be reported to Amcrest. They have fixed other API issues like this before.

---

calisro (Rob) 79 October 18, 2023, 2:17pm

> GaryOkie:
>
> If the firmware is configured for no chime, it probably won't trigger the chime kit relay.

I don't use the chime kit and never configured it. So should be good there.

> GaryOkie:
>
> Did you do a factory reset per the change log instructions? I'm not sure this is necessary, but it may be needed to ensure the configuration is correct.

I didn't at first but when I was having issues, I did and it didn't help. Still issues.

> GaryOkie:
>
> That will need to be reported to Amcrest. They have fixed other API issues like this before.

I saw some reddit posts and someone did report it to amcrest. Supposedly they're working on it. We'll see. I'll try again when a new firmware hits.

Since i've rolled back, things have calmed down again and seems to be working like before.

1 Like

---

flyinglow 80 October 18, 2023, 2:21pm

> calisro:
>
> Whenever the doorbell was rung, the amcrest integration went offline and I think the bell was restarting.

I kept having the same issue where the doorbell was restarting every time the bell was pressed.

I fixed it by replacing my Amcrest doorbell with the Reolink Wifi doorbell. It works great and the Reolink integration to Home Assistant has all the functions I need, like human detection and an event triggered when the bell is pressed.

---

calisro (Rob) 81 October 18, 2023, 2:27pm

> flyinglow:
>
> Reolink Wifi doorbell

That's not actually fixing it but it is getting rid of the problem. I've considered it. There are some things I do not like about the Amcrest. When this thing dies, i'll probably move to it. I've been following a few go2rtc threads on that camera. Most comments are positive.

Platonicsolid 82 January 30, 2024, 12:53pm

Morning!
So the integration is now entirely failing for me. There doesn't seem to be a way to get detailed logging, I'm not sure why, it still shows up in the Amcrest app and I still get the video feed.
Now I just get a "Error fetching camera details."
Since this integration isn't really maintained anymore, seems like I'm just SOL?

GaryOkie 83 January 30, 2024, 4:18pm

I assume you are referring to the custom Dahua integration. I did not know that RRoller had abandoned it, but it does look that way after reviewing his Github issues log.

The Amcrest integration is still supported as it is CORE, so switch to it and see if the problem persists for you. If it works, you will likely have to make adjustments in your automations for triggers on amcrest events as opposed to automatically created binary sensors. In my experience, the amcrest event trigger method has always been reliable for button press and human/motion detection.

---

carefulcomputer 84 February 4, 2024, 2:14am

I have latest firmware and have tons of problem. My door bell goes offline multiple times (reboots when pressing button), the state of ring/human detect gets stuck on 'on' all the time.
I want to try rolling back the firmware to older version. @calisro , would appreciate if you can share the older version. and also how to roll back if you have instructions.

Will really appreciate it.

Update: thanks to wayback machine, I found this link… **https://amcrest-firmwares.s3.amazonaws.com/Lexus_SDB61i-Lexus2_Eng_N_32M_nonCA_V1.000.00AC001.0.R.20230221.bin**

Update2: here are instructions to update the firmware manually - **https://support.amcrest.com/hc/en-us/articles/12358637857677-How-to-Upgrade-Downgrade-Amcrest-Smart-Home-Firmware-Manually-Using-the-Amcrest-IP-Config-Tool**

---

odwide 85 February 4, 2024, 2:41am

> carefulcomputer:
>
> reboots when pressing button

This is a known problem due to the battery not holding enough charge any more to power the camera for a few seconds. When the doorbell button is pressed all external power is briefly routed to the chime. Solution is to replace the battery by opening up the camera and soldering. Or simply buying a different camera.

1 Like

---

carefulcomputer 86 February 4, 2024, 2:43am

any links on how to replace the battery ?

---

odwide 87 February 4, 2024, 2:52am

Check out on the Amcrest forum what these guys did : **AD410 chime kit replacement - Page 3 - Amcrest Forum**

1 Like

GaryOkie 88 February 6, 2024, 5:42pm

This is exactly what I did to fix my AD410 which had a severely swollen battery from last summer's heat wave.

It's a rather delicate operation to clip the original tiny battery circuit and resolder onto the new battery, but certainly doable if you have any soldering skill.

I will also add that you can simply remove the battery along with the battery heater circuit and then bypass the external chime. The battery isn't needed at all with the chime disconnected. You could then send a custom mp3 doorbell ring sound or voice message to internal speakers upon a button press trigger - along with a phone text alert.

---

johnvpr (johnvpr) 89 March 1, 2024, 9:06pm

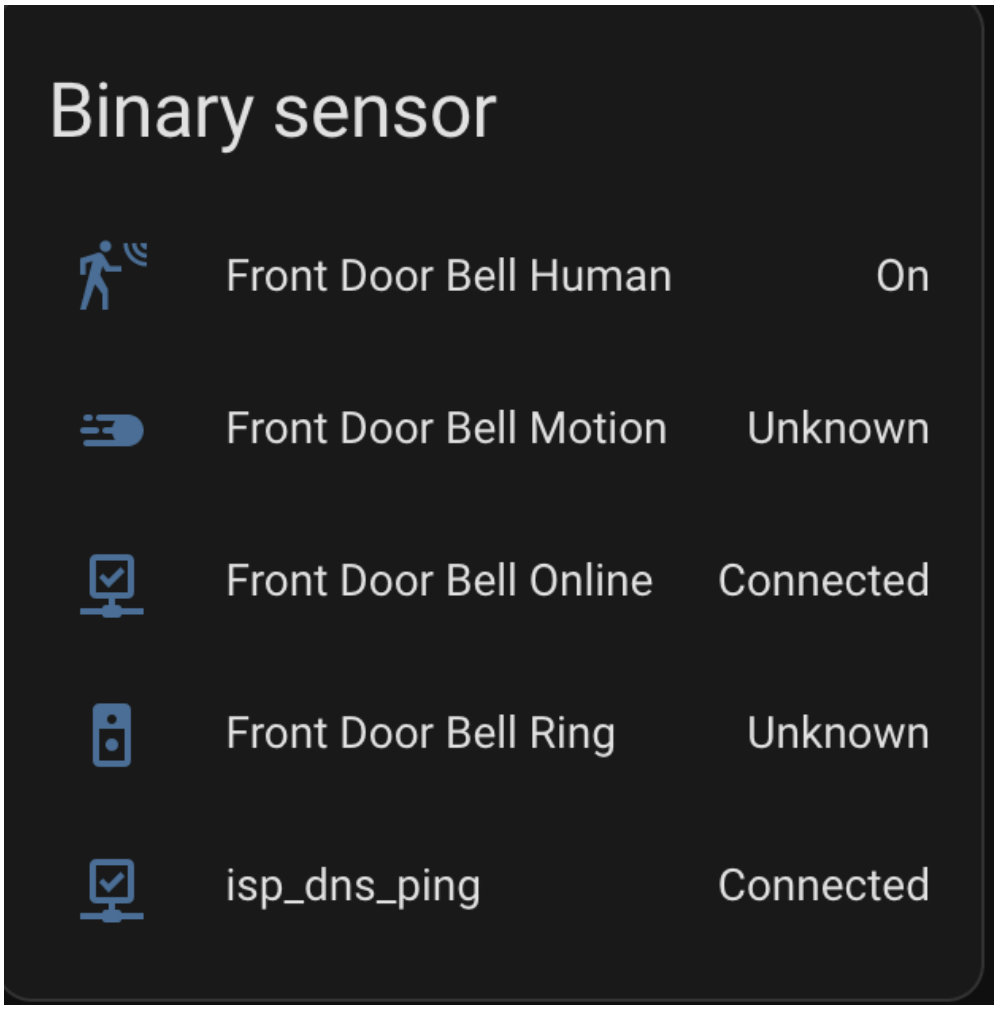
Guys, could someone post their working config, with an motion automation, I'm getting all the parts toghether but it seems not to work the automation.

Thanks!

---

johnvpr (johnvpr) 90 March 1, 2024, 9:11pm

Human works but ring and motion shows UNKNON



---

odwide 91 March 2, 2024, 4:07pm

If you are using ONVIF, sensors initially show Unknown and update next time the camera reports a state change.

johnvpr (johnvpr) 92 March 2, 2024, 9:00pm

@odwide So what is the best way to setup the camera?

---

odwide 93 March 2, 2024, 10:46pm

I have mine set through **ONVIF**, but using the **Amcrest integration** may be easier.

---

exx 94 March 3, 2024, 4:28pm

> Platonicsolid:
>
> Since this integration isn't really maintained anymore, seems like I'm just SOL?

Where did you read this?

---

odwide 95 March 3, 2024, 8:38pm

Just replaced the camera battery. Working fine again when the doorbell is pressed in cold weather. The old battery was swollen like a pillow.

GaryOkie 96 March 4, 2024, 4:17am

Looks good! I assume you didn't find an exact replacement battery but instead had to clip and solder.

---

odwide 97 March 4, 2024, 4:22am

Yes, exactly. I got the same model a guy over the Amcrest forums found.

LetThatSinkIn 98 March 6, 2024, 12:40am

Do you happen to know, can you just set it up to not use a chime at all from the get-go? My parents ordered one of these and I'm going to try and help the best I can over RDP to get things set up for them. They aren't going to need/use the chime portion at all, push notifications and a mp3 sound effect will be enough for a button push event. Just don't want to end up with the rebooting issues, if I can avoid it.

GaryOkie 99 March 6, 2024, 4:26am

Yes, during setup, you can choose no chime in the app, provided there is actually no chime connected.

RDP? You mean connect to a Windows PC remotely to set the AD410 up? Don 't see how that will work unless you have an Android or iOS emulator installed that will actually work with the SmartHome app - which is required for setup.

LetThatSinkIn 100 April 7, 2024, 10:28pm

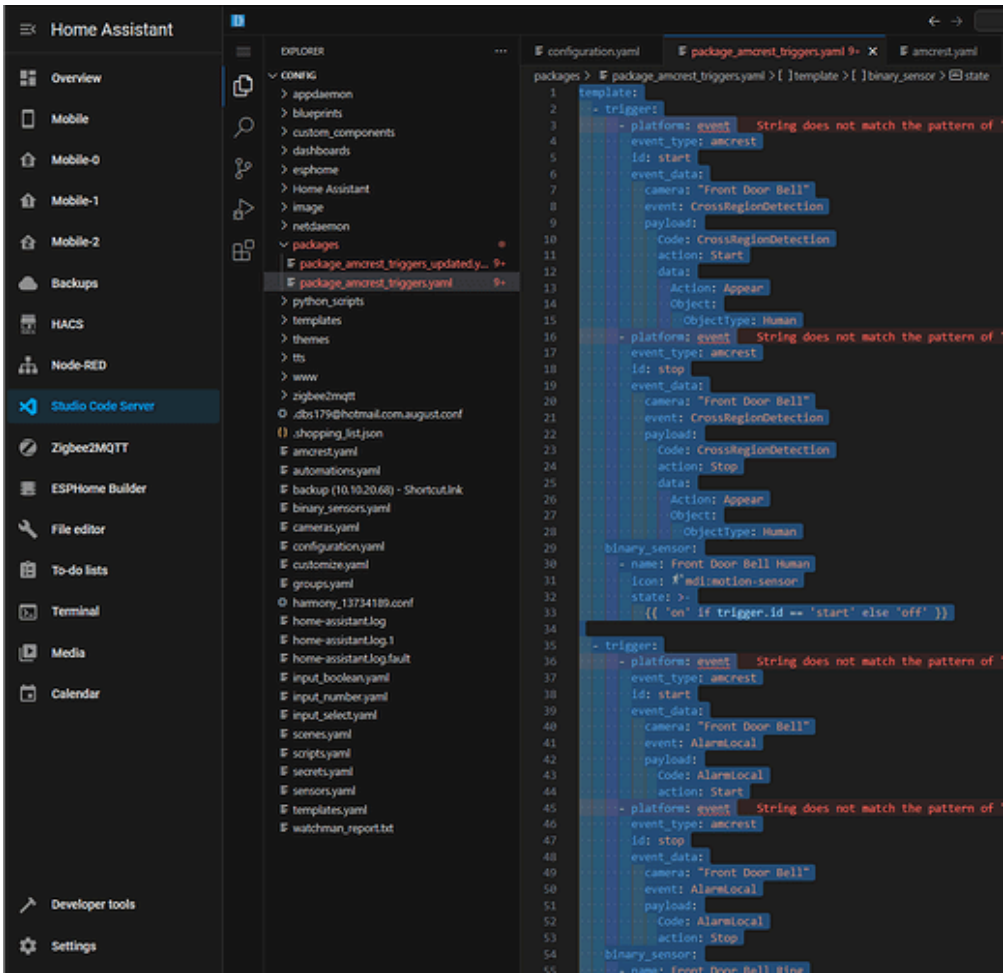
Got everything set up, no, I just meant that I could RDP into a windows machine down there to issue the URL based commands to tweak things with the camera. The setup was not smooth sailing, unfortunately. They only had iOS devices and I guess there are some known issues with the iOS app, probably would have been much easier if one of them had an Android. But in the end we got it working, and so far so good!

1 Like

---

dbs179 (dbs179) 101 June 18, 2025, 10:40pm

I know this is an old thread, but I'm in the middle of re-organizing all of my yaml so I just noticed this. If I look at the package_amcrest_triggers.yaml in studio code sever, I get the error "String does not match the pattern of "LEGACY_SYNTAX". Is there a way to restructure that yaml to get rid of the error?



GaryOkie 102 June 19, 2025, 3:03am

Try this:

> **String does not match the pattern of "LEGACY_SYNTAX^"**
>
> While editing automations.yaml with Studio Code Server it shows the message: String does not match the pattern of “LEGACY_SYNTAX^” next to lines like: platform: state platform: device Why is it complaining? How to fix? Update: I found that the legacy syntax issues will be fixed automatically, when the automation is (slightly) changed via the automation editor and then saved. The change then can be undone and the automation saved again.

1 Like

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dbs179 (dbs179) 103 June 19, 2025, 11:35am

Thank you! That did the trick.